SMART CAMPUS

MCA SemesterIV INDUSTRY PROJECT REPORT(MCA-41)

SUBMITTEDINPARTIALFULFILLMENTOFTHEREQUIREMENTSFORTHEAWARDOF THE DEGREE OF

MASTEROFCOMPUTER APPLICATIONS

SUBMITTEDTO

KADISARVAVISHWAVIDYALAYA

SUBMITTED BY

Nameof Student(s) UniversityExamNo(s)
Patel KrupalKumar S.

SUPERVISED BY
Mrs. Priyanka Shah –Internal
S K Patel Institute of Management and Computer Studies-MCA

SUPERVISED BY
Sagar Jasani
BrainyBeam Infotech pvt ltd

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Signature(s)of Students

Patel KrupalKumar S.

KadiSarvaVishwavidyalaya
S.K.PatelInstituteofManagementand ComputerStudies-MCA

CERTIFICATE FROM DIRECTOR

This is to certify that the project documentation entitled 'Smart Campus' is bonafide project work done by Mr. Patel KrupalKumar S, in partial fulfillment of the requirement for the award of degree of Master of Computer Application at S. K. Patel Institute of Management & Computer Studies-a constituent college of KADI SARVA VISHWAVIDYALAYA.

I further certify that the project work done by him is Original work carried out at BrainyBeam Infotech Pvt.Ltd Undertheir supervision and guidance.

Date:_____ *Director*

KadiSarvaVishwavidyalaya
S.K.PatelInstituteofManagementand ComputerStudies-MCA

CERTIFICATE FROM INTERNAL SUPERVISOR

This is to certify that the project report compiled by Mr. Patel KrupalKumar S. ,student of semester-4 of MCA department, S.K.Patel Institute of Management & ComputerStudies, a constituent college of Kadi Sarva Vishwavidyalaya, Gandhinagar,is done under my guidance. To the best of my knowledge, this work is satisfactorily completed. During the tenure, he was found to bevery hardworking and sincere. He completed all the tasks assigned and suggested to him.

Comments:

1

2

3

Date:_____

Name&signofInternal Supervisor

KadiSarvaVishwavidyalaya
S.K.PatelInstituteofManagementand ComputerStudies-MCA

DECLARATION

I here by declare that the project entitled "Smart Campus", submitted for the M.C.A. Degree is my original work and the project has not formed the basis for the Award of any degree or any other similar titles in any other University.

SignatureoftheStudent:

Place:

Date:

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1. Introduction of project:-

Project Title: Smart Campus System

The **Smart Campus System** is a web-based and IoT-enabled application developed to digitalise and simplify campus operations. It provides a centralized platform where student information, attendance, fees, and communication can be managed efficiently.

The system integrates multiple modules such as **student management, QR-based attendance, fee management, digital notices, and role-based access control**. By combining these functionalities, the system improves coordination between students, teachers, and administrators.

One of the major advantages of this system is that it reduces manual work and minimises human errors in attendance, record management, and fee tracking. Real-time data updates ensure accurate and up-to-date information across the campus.

Overall, the Smart Campus System enhances operational efficiency, improves communication, and provides a secure and organized environment for modern educational institutions.

2. Current / Existing System

- The traditional campus management system is commonly used in schools, colleges, and educational institutions where most of the daily activities are handled manually or with the help of basic digital tools such as registers, spreadsheets, and notice boards.

- In this system, different campus operations like **student record management, attendance tracking, fee collection, and communication** are managed separately without proper integration. For example, attendance is recorded manually in registers, student information is stored in files or Excel sheets, and fee records are maintained in separate ledgers or basic software.

- Communication between students, teachers, and administration is usually done through physical notice boards or verbal announcements, which may not reach all students effectively. This creates delays and miscommunication within the institution.

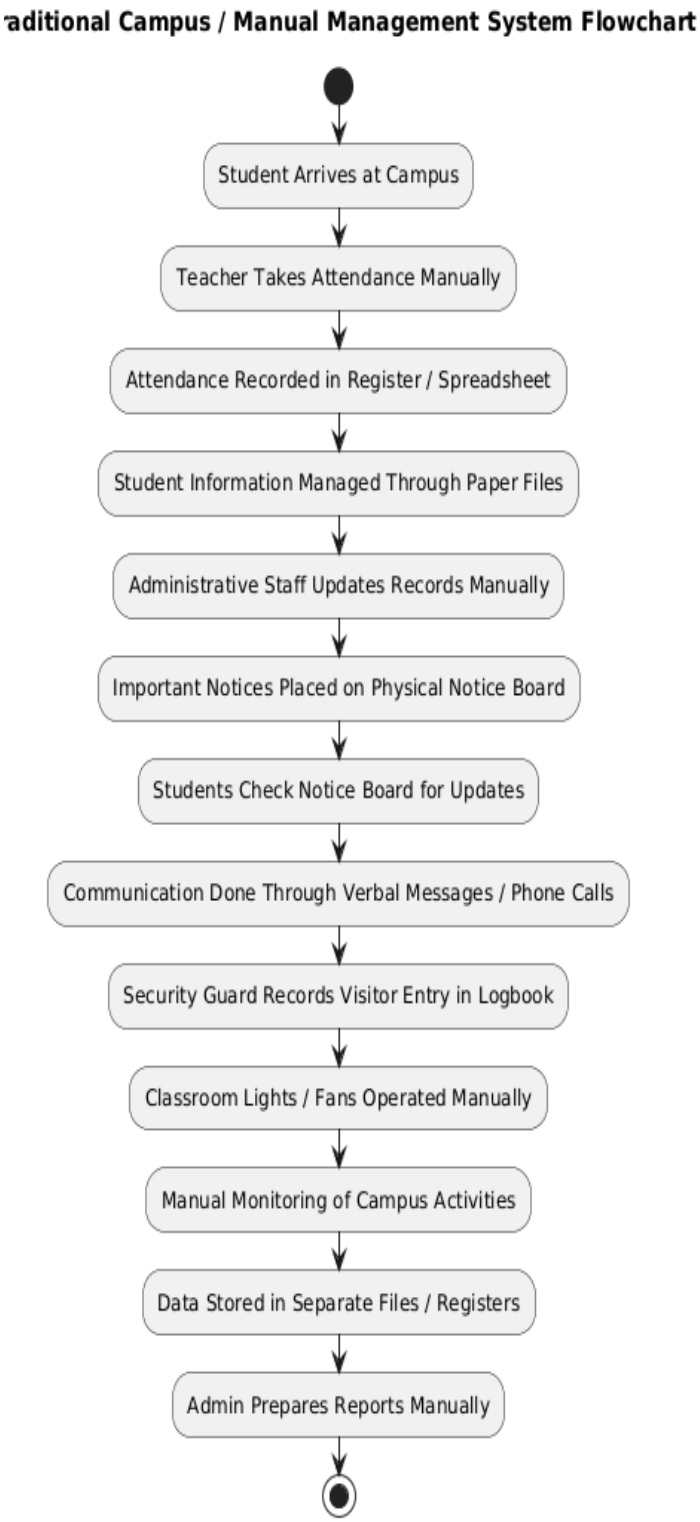
- Since there is no centralized system, data is scattered across multiple sources, making it difficult to access, update, and manage efficiently. The lack of automation increases the workload on staff and consumes a significant amount of time in performing routine tasks.

- Moreover, these systems do not support real-time updates. Any changes in attendance, fees, or student records must be manually updated, which increases the chances of errors and inconsistencies. As the number of students and campus activities grows, managing data becomes more complex and less reliable.

- Security is also a major concern in traditional systems, as there is no proper role-based access control. Sensitive data can be easily accessed or modified without authorisation.

- Overall, while traditional systems may be suitable for small-scale institutions, they become inefficient, time-consuming, and error-prone in larger campuses, highlighting the need for a smart and automated solution.

2.1 Flowchart of existing system



2.2 Drawbacks of Existing System:-

1. Lack of Real-Time Monitoring

In the traditional campus system, activities such as attendance marking, student record updates, and administrative data management are not performed in real time. Information is usually recorded manually and updated later, which creates delays in data availability.

2. High Dependency on Manual Processes

The existing system heavily relies on manual methods such as maintaining registers, writing reports, and updating spreadsheets. Tasks like attendance marking, record keeping, and communication require significant human effort.

3. Limited System Integration

In traditional systems, different campus functions such as student management, attendance tracking, fee management, and communication are handled separately. Each process operates independently without being connected to a centralised system.

4. Poor Communication System

Communication in traditional campuses is mainly done through physical notice boards, announcements, or word-of-mouth. This method is not reliable, as students may miss important information due to absence or lack of awareness.

5. Lack of Centralised Control

Campus operations in traditional systems are scattered across multiple registers, files, and basic software tools. There is no single platform where all activities can be monitored and controlled.

3.Proposed System:-

3.1 Modules

1. Admin Module

2. Teacher Module

3. Student Module

1. Admin Module

1. Admin can securely log in and manage their account.
2. The dashboard displays overall campus data such as total students, attendance status, and fee reports.
3. Admin can manage users such as teachers and students with proper roles and permissions.
4. Admin can add, update, and delete student and teacher records.
5. Attendance records can be monitored centrally.
6. Admin can manage fee details, payments, and pending dues.
7. Admin can post notices and announcements for students and staff.
8. Admin can generate reports such as attendance reports, fee reports, and performance reports.
9. Admin has full control over the entire system and campus operations.

2. Faculty Module

1. Teacher can securely log in and access system features based on permissions.
2. Teacher can mark attendance using QR-based system.
3. Teacher can view and manage student records.
4. Teacher can upload notices, assignments, and updates.
5. Teacher can monitor student attendance and academic performance.
6. Teacher can communicate with students through the system.
7. Teacher can access reports related to attendance and students.
8. Teacher helps in maintaining smooth academic activities.

3. Student Module

1. Student can securely log in and access their dashboard.
2. Students can mark attendance using QR codes.
3. Students can view their attendance records.
4. Students can check fee details and payment status.
5. Students can receive digital notices and updates.
6. Students can view personal and academic information.
7. Students can download receipts and reports.
8. Students can stay updated with campus activities.

3.2 Features of the Proposed System

1. **Secure Login and Authentication**
   The system provides secure login access for Admin, Teacher, and Student with role-based permissions.

2. **Dashboard Management**
   Displays important campus information such as student data, attendance, fees, and notices in one place.

3. **Student Management System**
   Allows managing student records such as personal details, academic data, and attendance.

4. **QR-Based Attendance System**
   Enables fast and accurate attendance tracking using QR codes.

5. **Fee Management System**
   Tracks fee payments, pending dues, and generates digital receipts.

6. **Digital Notice Board**
   Allows instant communication between admin, teachers, and students.

7. **Real-Time Data Management**
   Updates all data instantly to ensure accuracy and reliability.

8. **Report & Analysis**
   Generates reports such as attendance reports, fee reports, and performance analysis.

9. **Centralized Data System**
   All campus data is stored and managed in a single platform.

3.3 Scope of the Proposed System

1. The system is designed to manage student data, attendance, fees, and communication in a single integrated platform.
2. It allows the admin to monitor and control all campus activities efficiently.
3. Teachers can manage attendance, student performance, and academic updates.
4. Students can access their data, attendance, and notifications anytime.
5. The system provides real-time updates, improving data accuracy.
6. It supports report generation for better decision-making.
7. The system improves overall campus efficiency by reducing manual work.
8. It enhances communication and transparency within the institution.

4. Hardware and Software Specification

4.1 Hardware Requirement

- Processor: Intel Core i3 /i5 AMD Ryzen 3/5 or above
- RAM: 8 GB /16GB
- Storage: 256 GB SSD/512 GB SSD
- System Type: 64-bit system
- Display: Standard monitor with 1366 × 768 resolution or HD(1920 X 1080)
- Input Devices: Keyboard and Mouse , Optical barcode scanner and receipt printer for pos (OPTIONAL)
- Internet Connection: Required for package installation, updates, and deployment-related tasks

4.2 Software Requirement

- **Operating System:**
  Windows 10 / Windows 11 / Linux
- **Frontend:**
  HTML5, CSS3, JavaScript (Django Templates / Bootstrap optional)
- **Backend:**
  Python, Django Framework
- **Database:**
  SQLite (Default) / PostgreSQL
- **Tools:**
  VS Code, Git, GitHub, Postman
- **Browser:**
  Google Chrome / Edge / Firefox
- **Package Manager:**
  pip (Python Package Manager)

5. Analysis:-

5.1 Fact-finding techniques:-

Feature	Traditional Campus System	Basic Digital System	Smart Campus (Proposed System)
Attendance System	Manual register entry	Semi-digital (Excel)	QR-based real-time attendance
Data Management	Paper-based records	Stored in files/spreadsheets	Centralized digital database
Data Accuracy	Low (high errors)	Medium	High (real-time updates)
Centralized Control	Not available	Limited	Fully centralized admin control
Role-Based Access	Not defined	Basic access	Admin / Teacher / Student roles
Communication System	Notice board / verbal	Emails (limited)	Digital notice board (instant updates)
Fee Management	Manual records	Basic software	Fully automated fee tracking
Report Generation	Manual or delayed	Basic reports	Real-time reports & analytics
Student Management	Manual files	Limited digital records	Complete student management system
Real-Time Monitoring	Not available	Limited	Fully real-time system
Multi-User Support	Limited	Moderate	Fully supported
Data Security	Low	Medium	High (authentication system)
Automation Level	Low	Medium	High
System Integration	Not available	Partial	Fully integrated system
Audit / Logs	Not available	Limited	Full system activity tracking

5.2 Process Model:-

- The project begins with **requirement gathering**, where system needs such as student management, attendance, and fee handling are identified.
- The system is divided into modules like **Admin, Teacher, and Student** for better organization.
- Development is carried out in **small phases (sprints)**, where each feature is developed step by step.
- A **simple and user-friendly interface** is designed for easy interaction.
- Technologies like **HTML, CSS, JavaScript, and backend frameworks** are used for development.
- Features such as **QR attendance, fee management, and digital notices** are implemented gradually.
- Each module is **tested individually** to find and fix errors.
- After testing, the system is **deployed and used in real-time** campus scenarios.

5.3 Feasibility Study:-

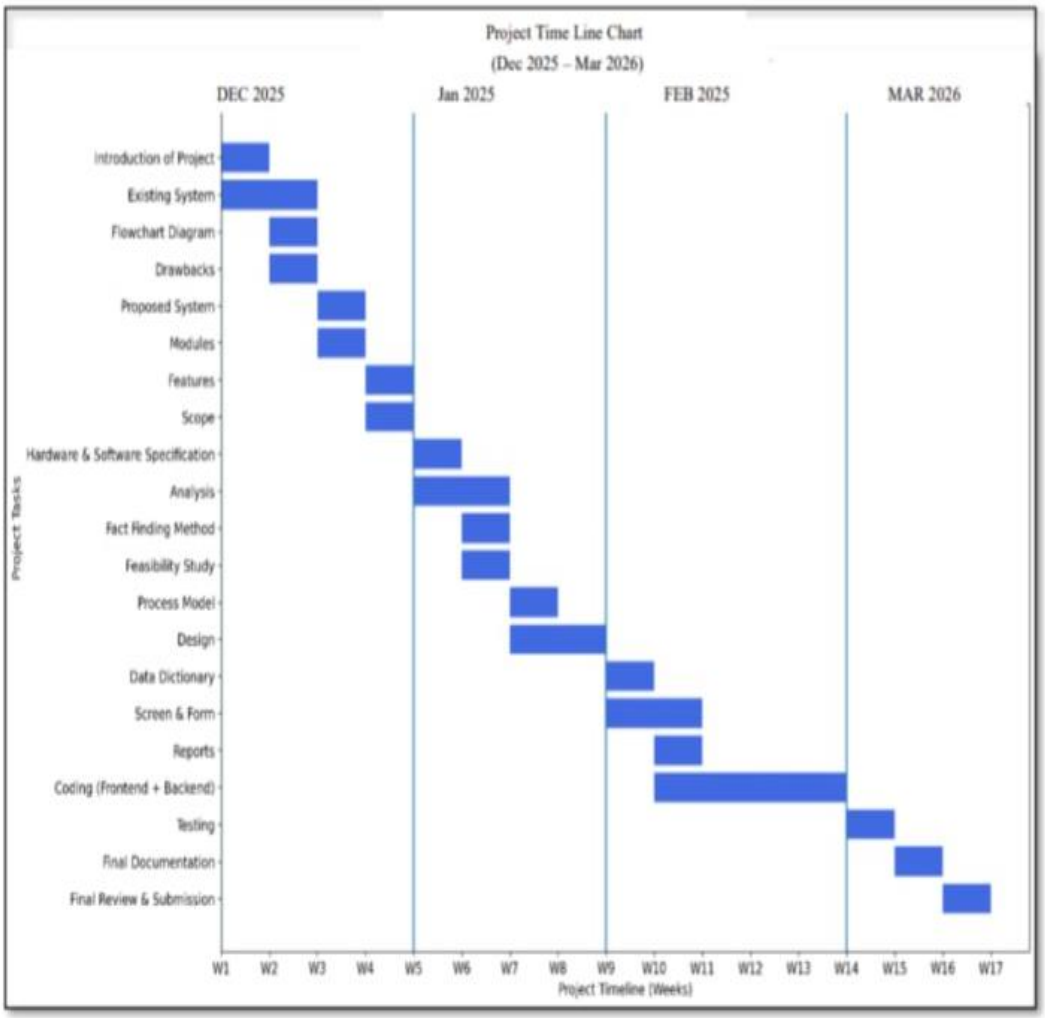
Technical Feasibility:
The system is technically feasible as it is developed using widely used technologies such as HTML, CSS, JavaScript for the frontend and Python with Django for the backend. The database (SQLite/PostgreSQL) is reliable and easy to manage. The hardware requirements are minimal (8GB RAM, standard processor), making it suitable for most institutions. Features like QR-based attendance can be implemented using existing tools and libraries, ensuring smooth development and deployment.

Economic Feasibility:
The project is cost-effective because it uses open-source technologies, eliminating licensing costs. Development tools like VS Code and GitHub are free to use. The system reduces manual work, saving time and operational costs in the long run. Overall, the benefits such as improved efficiency and reduced errors outweigh the initial development cost.

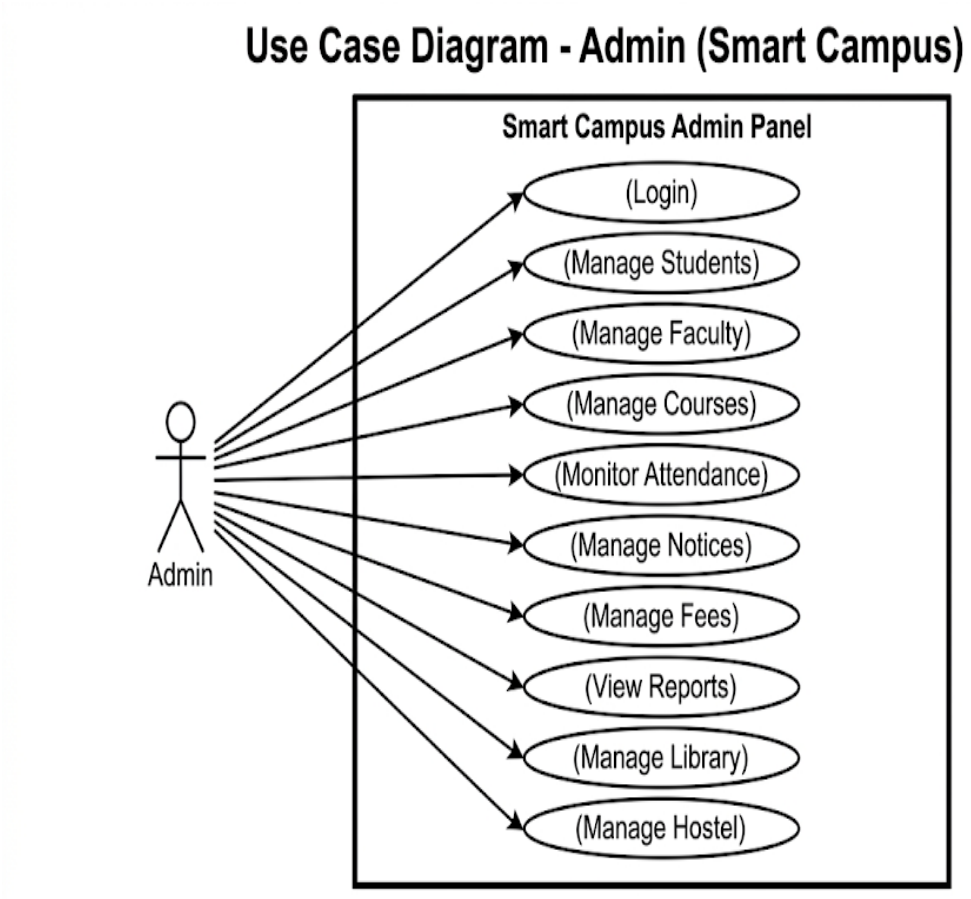
Operational Feasibility:
The system is designed to be user-friendly and easy to operate. Admins, teachers, and students can access the system through role-based dashboards. It simplifies attendance, fee management, and communication through a centralized platform. Minimal training is required, and it significantly reduces manual workload while improving accuracy and efficiency.

5.4 Diagrams

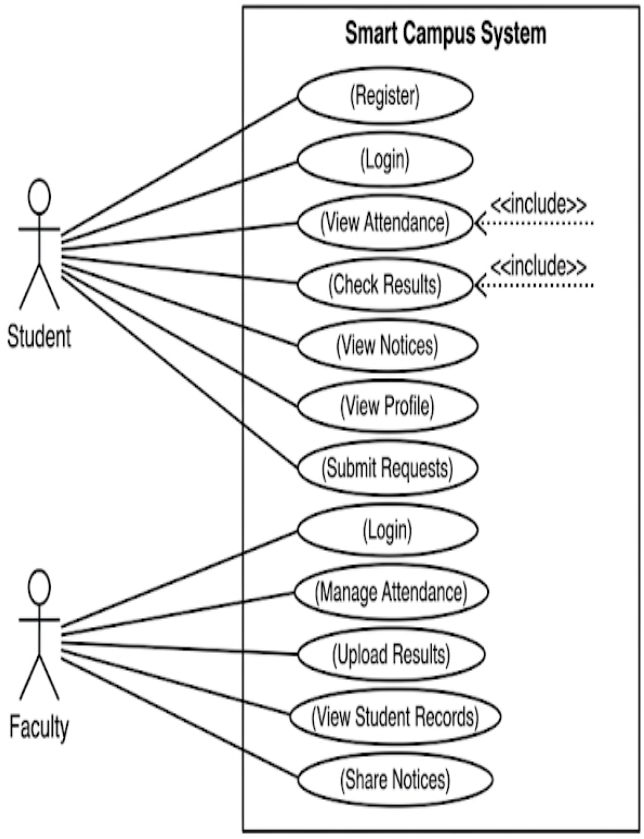
5.4.1 Estimated Gantt Chart/Time Line Chart:-



5.4.2 Use Case Diagram

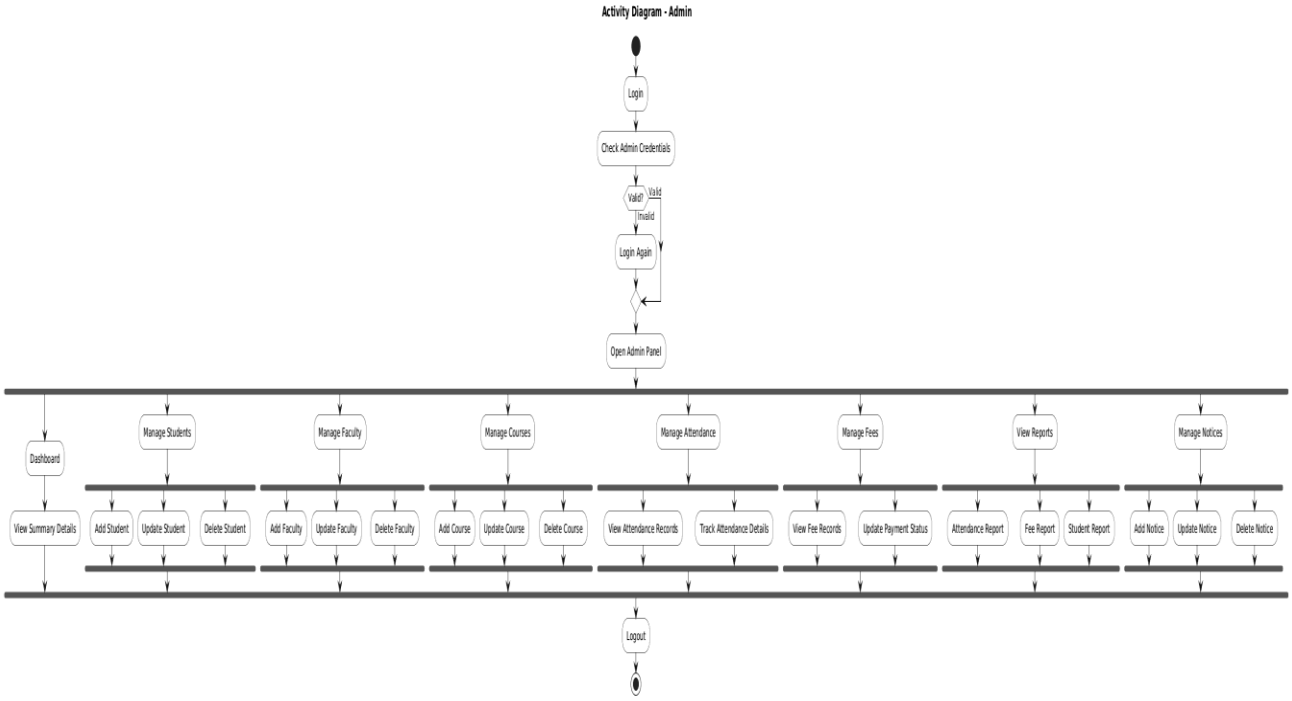


Use Case Diagram - Student / Faculty (Smart Campus)



II. Activity Diagrams:-

Admin

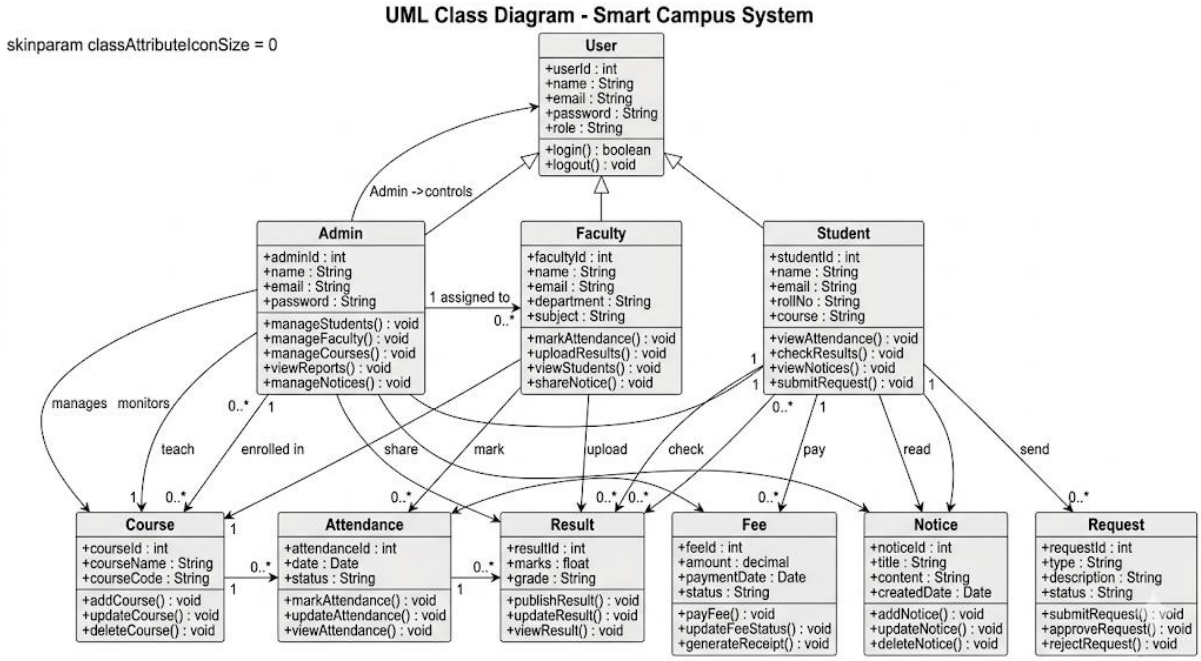


Student

Faculty

Activity Diagram - Faculty
Login
Validate Faculty Credentials
Valid?
Valid
Invalid
Login Again
Open Faculty Panel
Manage Attendance
Mark Attendance
Update Attendance
View Attendance Records
Manage Results
Enter Marks
Update Results
View Results
View Student Records
Access Student Details
Manage Subjects
View Assigned Courses
Share Notices
Create Notice
Update Notice
Handle Requests
View Student Requests
Approve / Reject Requests
Logout

III. CLASS DIAGRAM



6. Design:-

6.1 Data Dictionary:-

Table Name: Users

Field Name	Data Type	Description
user_id	INT (PK)	Unique ID of the user
name	VARCHAR(100)	Full name of the user
email	VARCHAR(100)	Email address of the user
password	VARCHAR(255)	Encrypted password
role	VARCHAR(50)	User role (Admin, Teacher, Student)
phone	VARCHAR(15)	Contact number
address	TEXT	User address
created_at	TIMESTAMP	Account creation date and time

Table Name: Courses

Field Name	Data Type	Description
course_id	INT (Primary Key)	Unique identifier for each course
course_name	VARCHAR(100)	Name of the course
course_code	VARCHAR(50)	Code assigned to the course

Table Name: Students

Field Name	Data Type	Description
student_id	INT (Primary Key)	Unique ID of the student
name	VARCHAR(150)	Full name of the student
roll_no	VARCHAR(50)	Roll number of the student
course_id	INT (Foreign Key)	Reference to course enrolled
email	VARCHAR(100)	Student email address
address	VARCHAR(255)	Student address
created_at	TIMESTAMP	Record creation date and time

Table Name: Faculty

Field Name	Data Type	Description
faculty_id	INT (PK)	Unique faculty ID
name	VARCHAR(150)	Faculty name
department	VARCHAR(100)	Department name
email	VARCHAR(100)	Email address
stock	INT	Available stock
image	VARCHAR(255)	Image file path
created_at	TIMESTAMP	Record creation time

Table Name: ATTENDANCE

Field Name	Data Type	Description
attendance_id	INT (PK)	Unique attendance ID
student_id	INT (FK)	Reference to student
course_id	INT (FK)	Reference to course
date	DATE	Attendance date
status	VARCHAR(20)	Attendance status (Present/Absent)

Table Name: Result

Field Name	Data Type	Description
result_id	INT (PK)	Unique result ID
student_id	INT (FK)	Reference to student
course_id	INT (FK)	Reference to course
marks	FLOAT	Marks obtained
grade	VARCHAR(20)	Grade

Table Name: Request

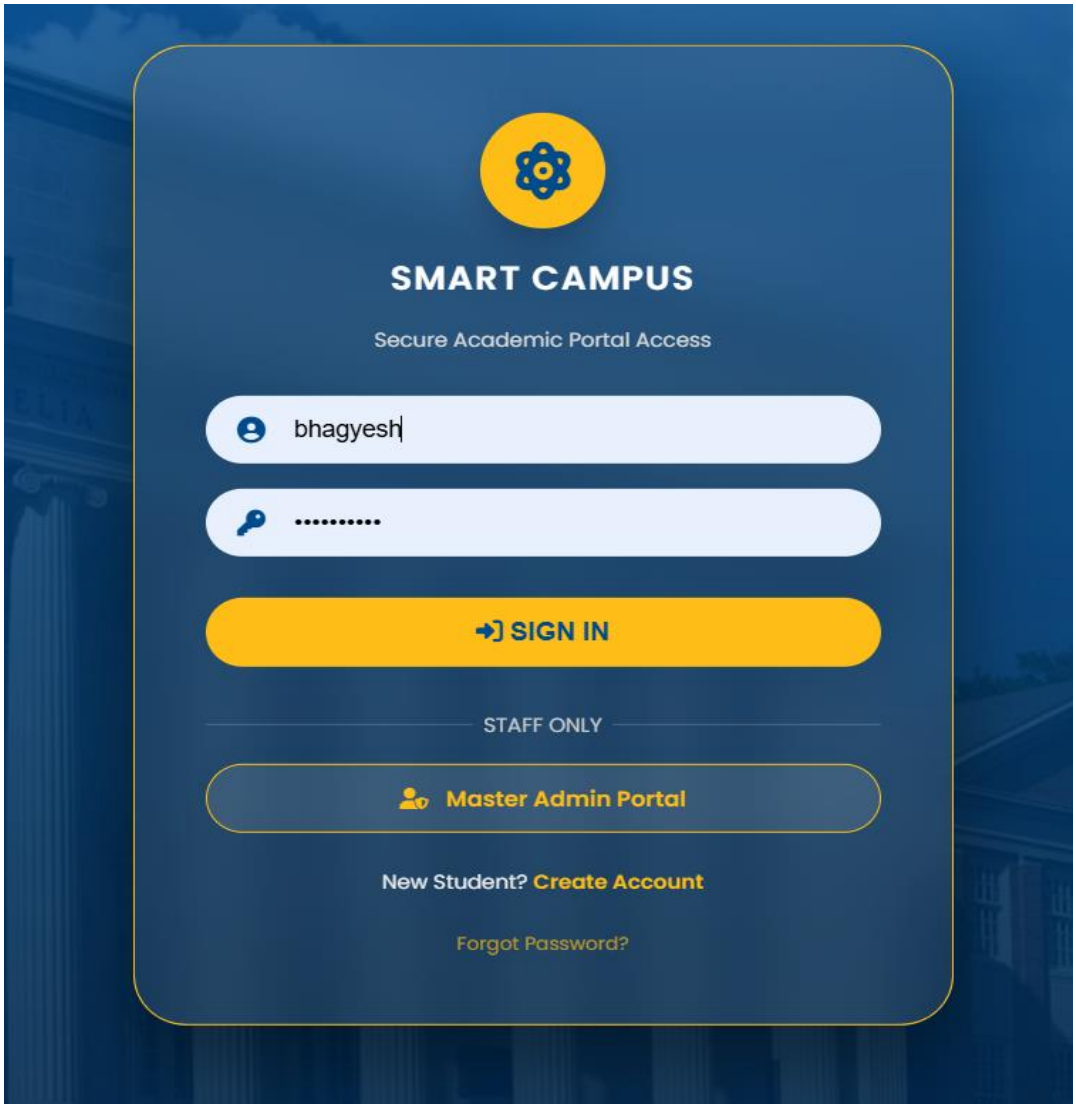
Field Name	Data Type	Description
request_id	INT (PK)	Unique request ID
student_id	INT (FK)	Reference to student
type	VARCHAR(100)	Request type (Leave/Complaint)
description	TEXT	Request details
status	VARCHAR(20)	Status (Pending/Approved/Rejected)

Table Name: Notice

Field Name	Data Type	Description
notice_id	INT (PK)	Unique notice ID
title	VARCHAR(150)	Notice title
content	TEXT	Notice content
created_at	TIMESTAMP	Creation time

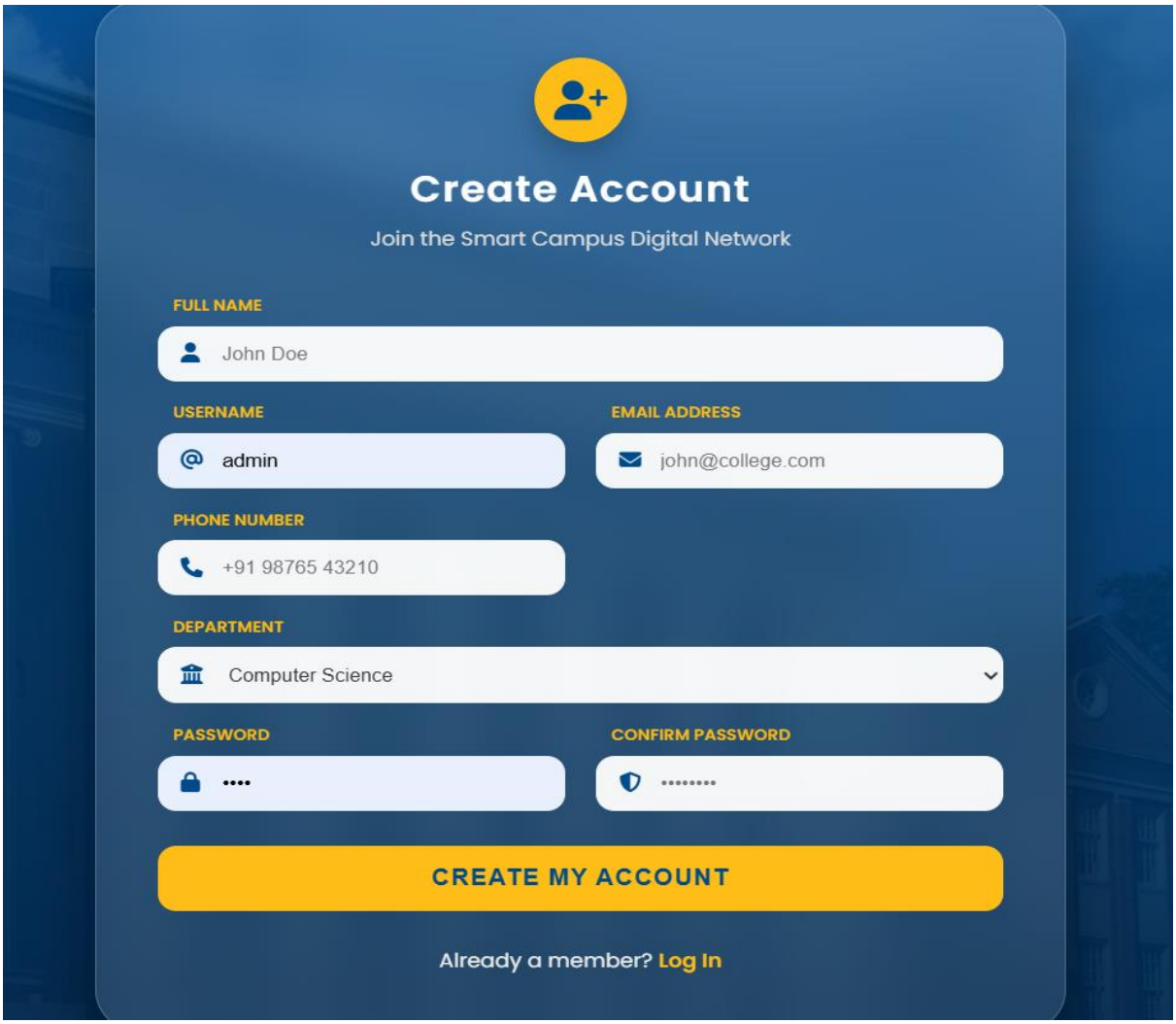
6.2 Screens/Forms:-

1) **Login page: -**



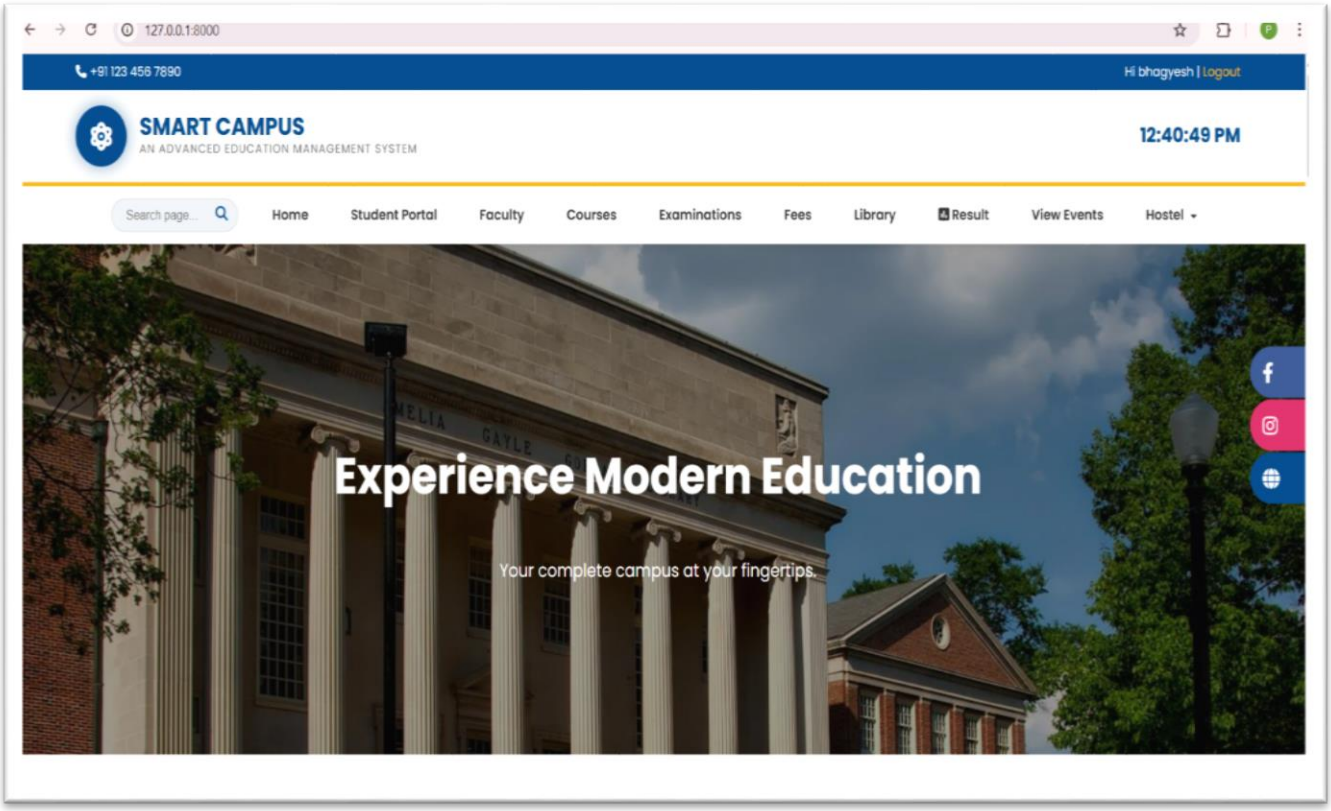
Secure login interface for students and staff to access the Smart Campus system using username and password.

2)Register page: -



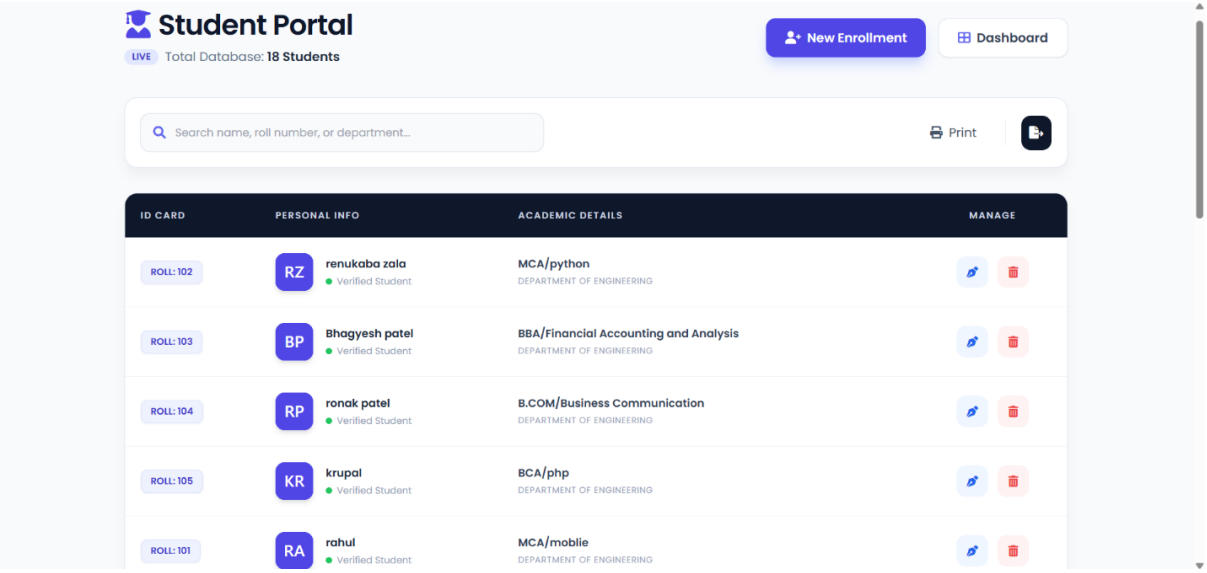
- Allows new students to register by entering personal, academic, and login details.
- Ensures secure account creation with password confirmation and department selection.

3)index page:-



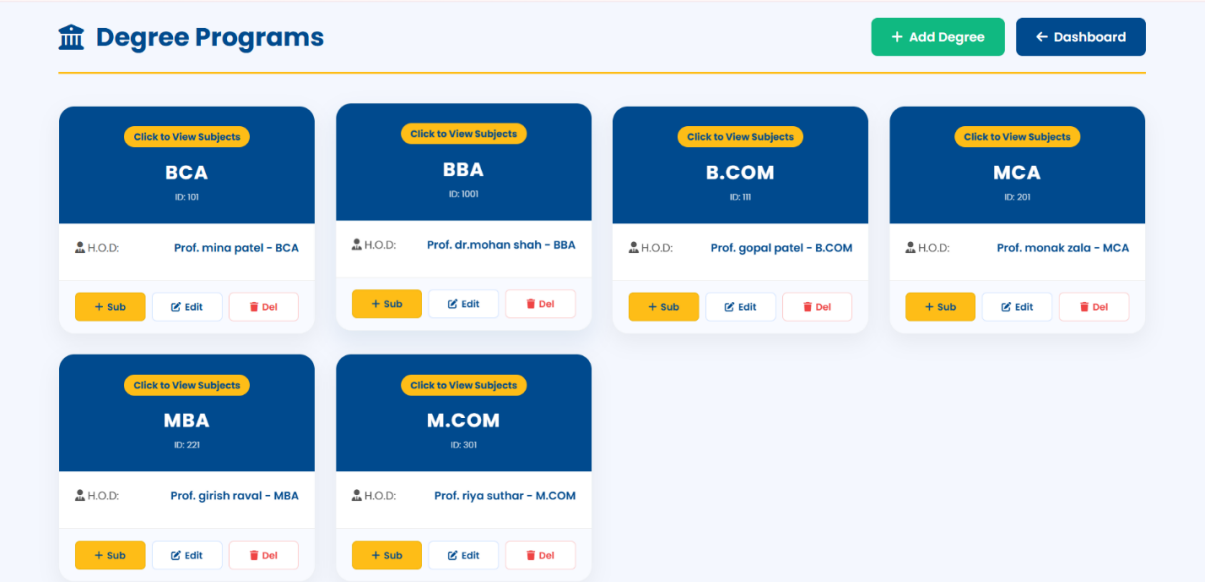
- Central dashboard providing navigation to modules like students, courses, exams, fees, and results.
- Displays a welcome interface with quick access menus and real-time system information.

4)student portal:-



- Displays student records including personal and academic details in a structured table format.
- Allows actions like edit, delete, print, and enrollment management for each student.

5) subject page:-



- Shows available degree programs with details like course name, ID, and assigned HOD.
- Enables management actions such as adding subjects, editing programs, and deleting entries.

6.3 Report:

The Smart Campus System **provides advanced** reporting and analytical features to help administrators, teachers, and management make informed decisions based on real-time data.

The system includes the following key reports:

Explanation: This module provides detailed insights into campus operations through reports and graphs. It helps in better decision-making, improves transparency, and enables data-driven management of academic and administrative activities.

7.Testing:-

Test ID	Module	Test Description	Input / Action	Expected Outcome	Result
TC_L01	Login	Valid login	Enter correct username & password	User successfully logged in	Pass
TC_L02	Login	Invalid login	Wrong password	Error message displayed	Pass
TC_L03	Login	Empty fields	Click login without input	Validation message shown	Pass
TC_L04	Login	Forgot password	Click "Forgot Password"	Redirect to reset page	Pass
TC_R01	Registration	Valid registration	Fill all fields correctly	Account created successfully	Pass
TC_R02	Registration	Missing fields	Leave required fields empty	Validation errors displayed	Pass
TC_R03	Registration	Password mismatch	Enter different passwords	Error message shown	Pass
TC_R04	Registration	Invalid email	Enter wrong email format	Email validation error	Pass
TC_D01	Dashboard	Page load	Login and open dashboard	Dashboard loads successfully	Pass
TC_D02	Dashboard	Navigation menu	Click menu items	Correct page opens	Pass
TC_D03	Dashboard	Logout	Click logout	User logged out successfully	Pass
TC_D04	Dashboard	Search	Enter keyword in search	Relevant results displayed	Pass
TC_S01	Student Portal	View students	Open student portal	Student list displayed	Pass

Test ID	Module	Test Description	Input / Action	Expected Outcome	Result
TC_S02	Student Portal	Search student	Enter name/roll no	Filtered results shown	Pass
TC_S03	Student Portal	Add student	Click "New Enrollment"	Form opens successfully	Pass
TC_S04	Student Portal	Edit student	Click edit button	Student data updated	Pass
TC_S05	Student Portal	Delete student	Click delete button	Student removed successfully	Pass

8. Future Enhancement:-

1. Online Fee Payment Integration

The Smart Campus System can be enhanced by integrating an **online fee payment system** that allows students and parents to pay fees digitally. This feature supports multiple payment methods such as **UPI, debit cards, credit cards, and secure payment gateways**.

2. Cloud Deployment

The system can be deployed on **cloud platforms** to improve scalability, performance, and accessibility. Cloud deployment allows users to access the system from anywhere at any time using the internet.

3. Multi-Campus Support

An advanced version of the system can support **multiple campuses or departments** within a single platform. This feature allows institutions with multiple branches to manage all operations centrally.

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4. Mobile Application

A mobile application can be developed to provide easy access to the system for students and faculty. Through the app, users can:

- Check attendance
- View results and performance
- Receive notices and updates
- Submit requests or queries

This ensures **anytime, anywhere access**, improving user experience and engagement.

5. AI-Based Student Analytics

Artificial Intelligence can be integrated to analyse student data such as attendance, academic performance, and behaviour patterns.

The system can generate **insights, predictions, and suggestions** to help teachers and administrators identify weak areas and improve student outcomes. For example:

- Predicting student performance
- Identifying irregular attendance
- Suggesting improvement strategies

This helps in **data-driven decision-making** and enhances overall educational quality.

9. References / Bibliography

Existing / Reference Systems:

- https://www.odoo.com/
- https://www.zoho.com/
- https://www.hubspot.com/products/crm
- https://www.fedena.com/ *(School Management System)*
- https://www.entab.in/ *(Campus ERP System)*

Our Inspiration / Tools Used:

- https://app.diagrams.net/ *(For diagrams and system design)*
- https://react.dev/ *(Frontend development)*
- https://vitejs.dev/ *(Build tool for fast development)*
- https://expressjs.com/ *(Backend framework)*
- https://nodejs.org/ *(Server-side runtime)*
- https://www.prisma.io/ *(Database ORM tool)*
- https://www.postgresql.org/ *(Database system)*
- https://tailwindcss.com/ *(UI styling framework)*
- https://developer.mozilla.org/ *(Web development reference)*